

Cube Transfer for Randomized Dvoretzky in ℓ_p^n

An ε -dependent logarithmic regime and a uniform threshold improvement

Candidate partial result for arXiv:1510.07284
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9 August 2026

Status. Candidate strong partial result, likely valid, pending specialist human review. The sharp $\varepsilon/\log(1/\varepsilon)$ upper bound is proved when $p \geq C\varepsilon^{-1}\log n$, and the source's uniform threshold $p > (\log n)^2$ is improved to $p \geq C(\log n)^2/\log \log n$ for nontrivial dimensions $k \geq 2$. The full logarithmic- p , very-small-distortion wedge remains open.

1 Source question

Paouris, Valettas and Zinn study the largest dimension of a Haar-random subspace of ℓ_p^n on which the norm is $(1 + \varepsilon)$ -spherical [1]. In Section 6, item 5, they ask whether the factor $\log(1/\varepsilon)$ in the denominator of

$$k \lesssim \frac{\varepsilon \log n}{\log(1/\varepsilon)}$$

is necessary throughout the range $p \gtrsim \log n$. Their Proposition 6.2 proves this upper bound for $p > (\log n)^2$.

5. The existence of $\log(1/\varepsilon)$ as $p \rightarrow \infty$. Note that Theorem 4.9 and furthermore Corollary 4.10 suggest that the concentration of the ℓ_p norm with $p \gtrsim \log n$ is similar with the one we get for the ℓ_∞ norm. This means that the classical net argument yields random subspaces which are $(1 + \varepsilon)$ -spherical as long as $k \lesssim \varepsilon \log n / \log \frac{1}{\varepsilon}$. We do not know if this $\log(1/\varepsilon)$ term is needed, for this range of p . As an easy corollary of the main result of [34], we have:

Proposition 6.2. *Let $p > (\log n)^2$ and $\varepsilon \in (0, 1/3)$. If the random k -dimensional subspace of ℓ_p^n is $(1 + \varepsilon)$ -spherical with probability greater than $3/4$, then $k \leq C\varepsilon \log n / \log \frac{1}{\varepsilon}$, where $C > 0$ is an absolute constant.*

Figure 1: The open question and Proposition 6.2 in arXiv:1510.07284, PDF page 39.

The present packet improves the known parameter range but does not close the whole question.

2 Definitions and external input

For a subspace $E \subset \mathbb{R}^n$ define its ℓ_q spherical distortion by

$$D_q(E) := \frac{\sup_{x \in E \cap S^{n-1}} \|x\|_q}{\inf_{x \in E \cap S^{n-1}} \|x\|_q}.$$

Thus E is $(1 + \eta)$ -spherical in ℓ_q^n precisely when $D_q(E) \leq 1 + \eta$.

We use the sharp randomized cube theorem of Tikhomirov [2]: there is an absolute constant C_T such that, if $k > 1$, $0 < \eta < 1/2$, and a majority of Haar-random k -subspaces of $\mathbb{R}^n$ satisfy $D_\infty(E) \leq 1 + \eta$, then

$$k \leq C_T \frac{\eta \log n}{\log(1/\eta)}. \quad (1)$$

The probability threshold $3/4$ used below is stronger than “a majority.”

3 Proof intuition

The key idea is to transfer a good ℓ_p section to the cube without changing the random subspace. Globally on $\mathbb{R}^n$ one has $\|x\|_\infty \leq \|x\|_p \leq n^{1/p} \|x\|_\infty$. On a fixed Euclidean sphere this implies

$$D_\infty(E) \leq n^{1/p} D_p(E).$$

Hence a $(1 + \varepsilon)$ -spherical ℓ_p section is a $(1 + \delta)$ -spherical cube section with $1 + \delta = (1 + \varepsilon)n^{1/p}$. Tikhomirov’s cube obstruction then applies verbatim. When $(\log n)/p$ is at most a constant multiple of ε , this has the desired sharp dependence on ε . Uniformly in ε , the same bound excludes every $k \geq 2$ once $p \gtrsim (\log n)^2 / \log \log n$ and ε lies below the transfer error.

4 Main transfer theorem

Theorem 1 (Cube-transfer upper bound). *Let $n > 1$, $1 \leq p < \infty$, $k > 1$, and $0 < \varepsilon < 1$. Set*

$$\delta := (1 + \varepsilon)n^{1/p} - 1. \quad (2)$$

Suppose $0 < \delta < 1/2$ and that a Haar-random k -dimensional subspace E of $\mathbb{R}^n$ satisfies $D_p(E) \leq 1 + \varepsilon$ with probability greater than $3/4$. Then

$$k \leq C_T \frac{\delta \log n}{\log(1/\delta)}, \quad (3)$$

where C_T is the absolute constant in the randomized cube theorem.

Proof. For every $x \in \mathbb{R}^n$,

$$\|x\|_\infty \leq \|x\|_p \leq n^{1/p} \|x\|_\infty. \quad (4)$$

Fix a subspace E . The first inequality gives

$$\sup_{E \cap S^{n-1}} \|x\|_\infty \leq \sup_{E \cap S^{n-1}} \|x\|_p,$$

whereas the second gives

$$\inf_{E \cap S^{n-1}} \|x\|_\infty \geq n^{-1/p} \inf_{E \cap S^{n-1}} \|x\|_p.$$

Dividing yields

$$D_\infty(E) \leq n^{1/p} D_p(E). \quad (5)$$

Consequently, every E with $D_p(E) \leq 1 + \varepsilon$ satisfies

$$D_\infty(E) \leq n^{1/p}(1 + \varepsilon) = 1 + \delta.$$

Thus more than $3/4$ of the Haar-random k -subspaces are $(1 + \delta)$ -spherical in the cube. Equation (3) is exactly (1) with $\eta = \delta$. $\square$

5 Consequences for the source question

Corollary 2 (ε -dependent logarithmic regime). *There is an absolute constant C such that the following holds. Let $0 < \varepsilon < 1/3$ and*

$$p \geq 4\varepsilon^{-1} \log n.$$

If a Haar-random k -subspace of ℓ_p^n is $(1 + \varepsilon)$ -spherical with probability greater than $3/4$, then

$$k \leq C \frac{\varepsilon \log n}{\log(1/\varepsilon)}. \quad (6)$$

Proof. Put $s = (\log n)/p$, so $s \leq \varepsilon/4$ and $\delta = (1 + \varepsilon)e^s - 1$. For $0 < \varepsilon < 1/3$, elementary estimates on the exponential give $\delta < 1/2$ and $\delta \leq (3/2)\varepsilon$. On this interval,

$$\frac{\delta}{\log(1/\delta)} \leq C_0 \frac{\varepsilon}{\log(1/\varepsilon)}$$

for an absolute C_0 . Theorem 1 proves (6). $\square$

In particular, for every fixed distortion $\varepsilon > 0$, the logarithmic loss is necessary already throughout $p \geq C_\varepsilon \log n$, rather than only for $p > (\log n)^2$.

Corollary 3 (Uniform threshold improvement). *There are absolute constants $A, C > 0$ and n_0 such that, for $n \geq n_0$, $k \geq 2$, $0 < \varepsilon < 1/3$, and*

$$p \geq A \frac{(\log n)^2}{\log \log n},$$

the probability hypothesis of Corollary 2 again implies (6).

Proof. Write $L = \log n$ and $s = L/p$. Then

$$s \leq \frac{\log L}{AL}. \quad (7)$$

If $\varepsilon \geq s$, then $\delta = (1 + \varepsilon)e^s - 1 < 1/2$ for n_0 large enough, and $\delta \leq C_1\varepsilon$. The same monotonicity comparison used in Corollary 2 gives (6).

Suppose instead that $\varepsilon < s$. For n_0 large, (7) gives $s < 1/16$, and

$$\delta = (1 + \varepsilon)e^s - 1 \leq (1 + s)e^s - 1 \leq 4s < 1/2.$$

Theorem 1 and monotonicity of $t \mapsto t/\log(1/t)$ give

$$k \leq C_2 \frac{sL}{\log(1/s)} \leq \frac{C_2}{A} \frac{\log L}{\log(AL/\log L)} \leq \frac{2C_2}{A}.$$

Choose the absolute constant A so large that $2C_2/A < 2$. This contradicts $k \geq 2$. Hence the probability hypothesis cannot hold in the case $\varepsilon < s$, and the remaining case already gives (6). $\square$

6 Verification and limitations

The proof has one external dependency: the sharp randomized cube upper bound (1). The transfer inequality (5) is exact and pointwise in the subspace, so it loses neither probability nor dimension. The estimates in the two corollaries only compare the increasing function $t/\log(1/t)$ at arguments differing by absolute factors.

The result does *not* settle the source question when

$$p \asymp \log n, \quad \varepsilon \ll \frac{\log n}{p}.$$

In this wedge the global comparison error $n^{1/p} - 1$ is larger than the desired distortion. A full solution appears to require a genuinely subspace-sensitive analogue of the cube anti-concentration argument, rather than a global norm comparison.

7 Novelty check and human review

A bounded search on 9 August 2026 covered the run’s lightweight indexes, the local arXiv source corpus, exact web/arXiv queries for the source title and the logarithmic- p question, and the OpenAlex citation graph of the source DOI (24 citing records). Lytova and Tikhomirov explicitly observed in 2017 that the randomized Dvoretzky problem was not sharp for logarithmic p [3]; their later result resolves the Gaussian variance transition, not this section-dimension question. No later full resolution or this exact two-parameter transfer statement was found. This bounded search does not prove novelty.

Recommendation: high-priority specialist verification. The proof is short. A reviewer should confirm the exact normalization of Tikhomirov’s cube theorem and the use of its “majority” hypothesis, then check the integer $k \geq 2$ exclusion in Corollary 3.

References

- [1] G. Paouris, P. Valettas and J. Zinn, *Random version of Dvoretzky’s theorem in ℓ_p^n* , Stochastic Process. Appl. **127** (2017), 3187–3227; arXiv:1510.07284.
- [2] K. E. Tikhomirov, *The randomized Dvoretzky’s theorem in ℓ_∞^n and the χ -distribution*, Geometric Aspects of Functional Analysis, Lecture Notes in Math. 2116, Springer (2014), 455–463.
- [3] A. Lytova and K. Tikhomirov, *The variance of the ℓ_p^n -norm of the Gaussian vector, and Dvoretzky’s theorem*, Algebra i Analiz **30** (2018), 107–139; arXiv:1705.05052.